PROJECT PRESENTATION

Prediction of Crop Yield using Regression Analysis



Rainfall, cultivated area and food price index as predictors of rice yield

Based on V. Sellam and E. Poovammal, Indian Journal of Science and Technology, Vol 9(38), October 2016

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Abstract

Goal: show how rainfall (AR), cultivated area (AUC) and food price index (FPI) shape rice yield.

Method: linear regression, with yield as the response variable.

Result: yield depends mainly on rainfall; area and price also influence it.

Next: extend with MSP, CPI and WPI.

3
factors: AR, AUC, FPI

1
crop studied: rice

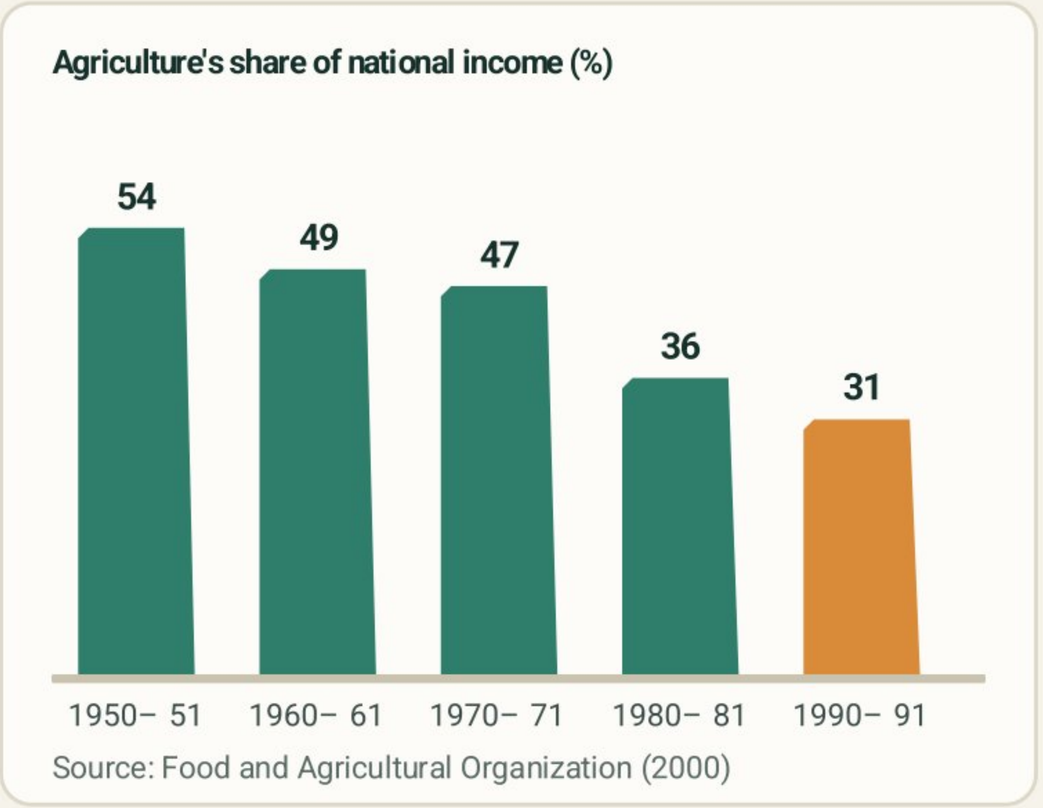
~10
years of data

0.72
R^2 for rainfall and area

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Introduction

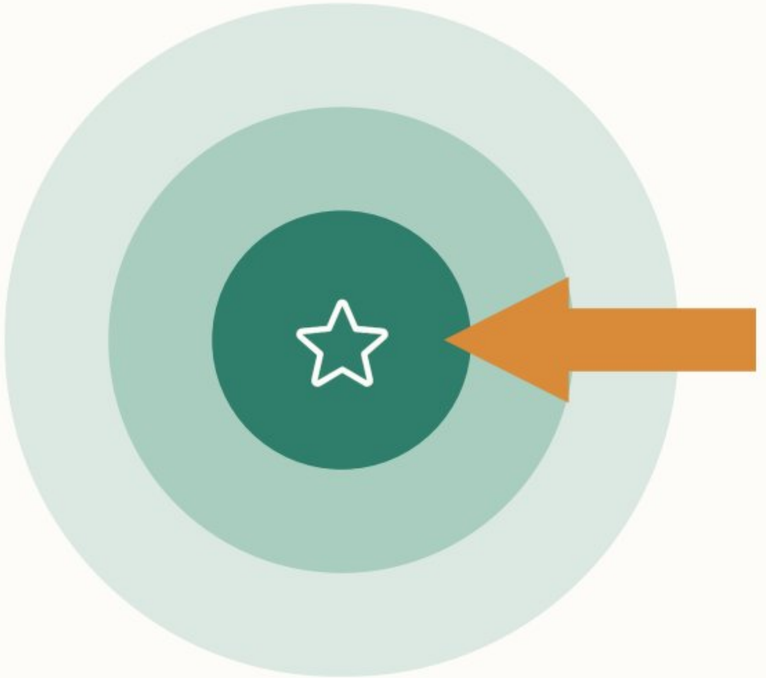
- Agriculture is a vital source of livelihood and employment in India.
- Its share of national income fell from 54% in 1950– 51 to 31% in 1990– 91.
- Weather uncertainty, soil nutrients, fertilizer cost and pests make yield hard to predict.
- Regression analysis can relate yield to rainfall, area and price.



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Objective

- Analyze how Annual Rainfall (AR), Area under Cultivation (AUC) and Food Price Index (FPI) affect crop yield.
- Establish the relationship among these parameters using regression analysis.
- Help farmers reduce losses and get better prices for their crops.
- Use rice as the case study, the most common crop in many areas of India.

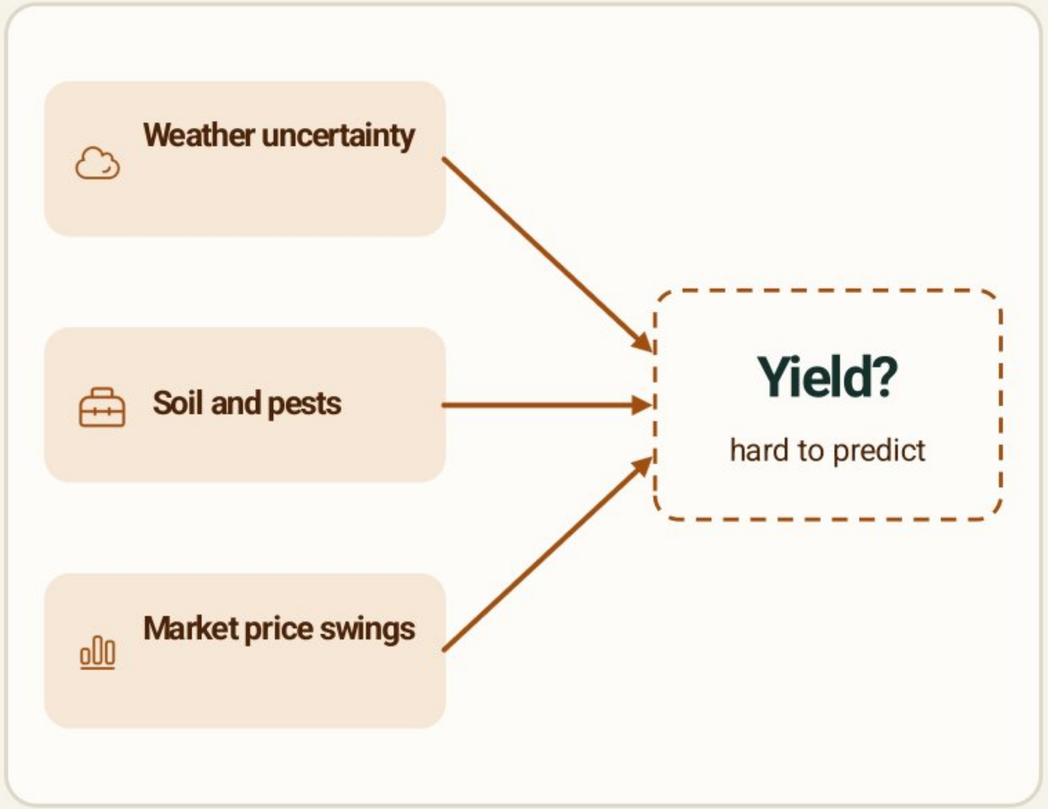


Predict rice yield from AR, AUC and FPI

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Existing System

- Yield is estimated from experience and seasonal guesswork.
- Weather, soil nutrients, fertilizer cost and pests lower output unpredictably.
- Earlier studies use pattern classification (k-means, SVM) and cropping-zone indices (RSI, RYI).
- Little quantified view of how rainfall, area and price drive rice yield together.



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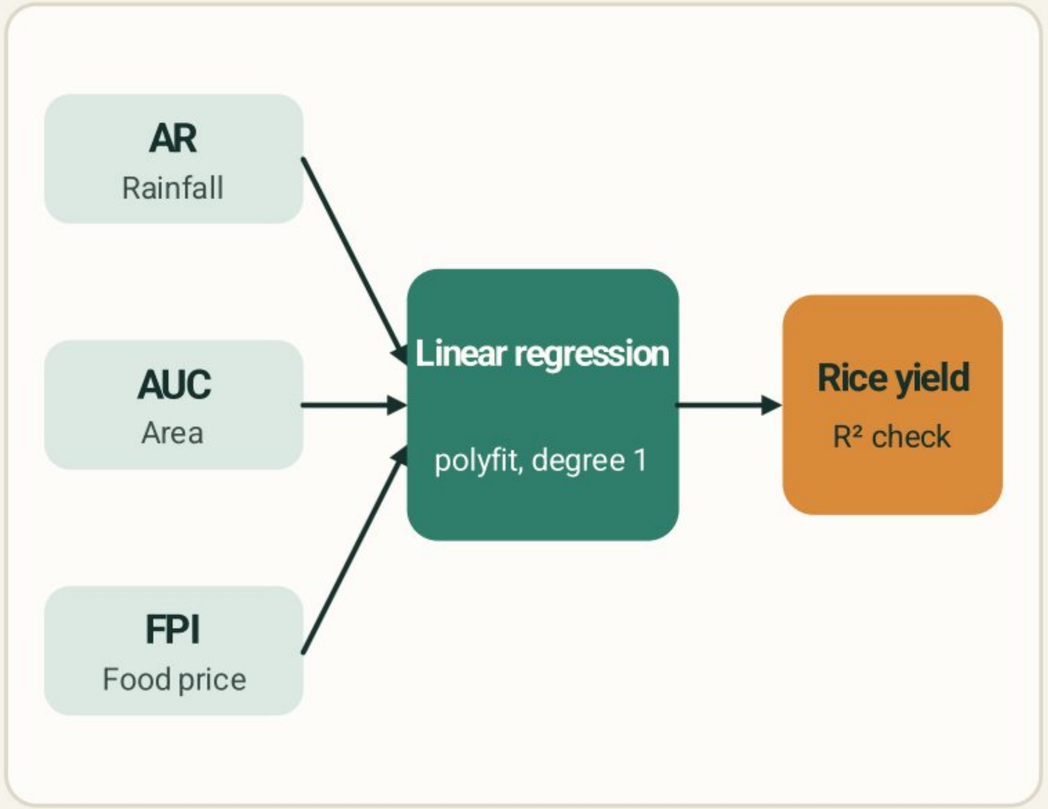
Proposed System

Linear regression of rice yield on rainfall (AR), area (AUC) and food price index (FPI), one factor at a time.

Built in MATLAB with a degree-1 polyfit and checked with R^2.

Four stages: build the model, compute residuals, compute residual sum of squares and R^2, then implement on the data.

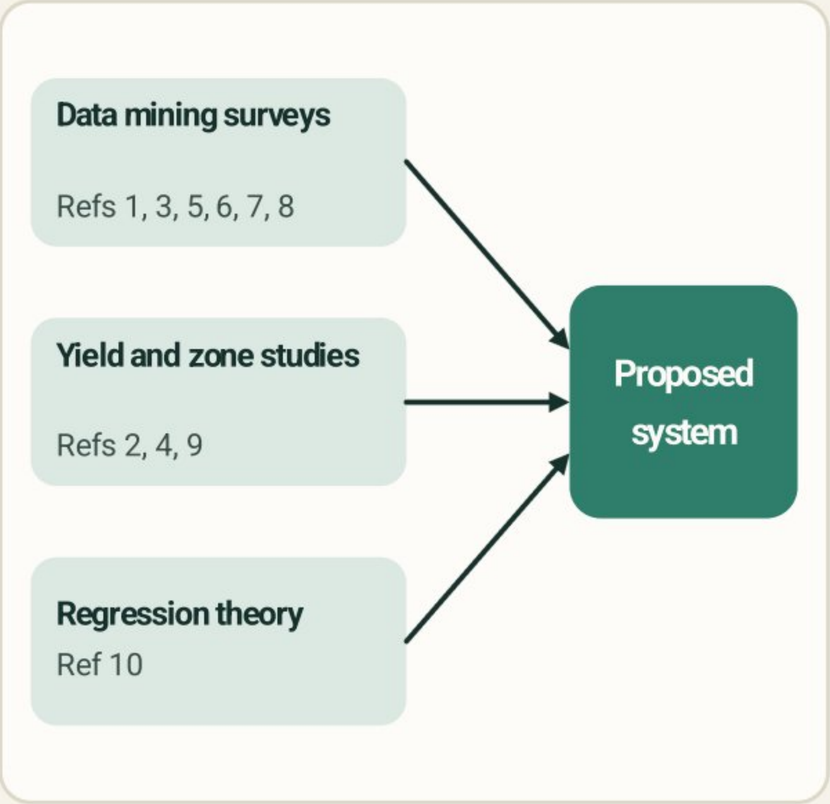
An R^2 above 0.5 is read as a strong link to yield.



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Proposed System: Reference Papers

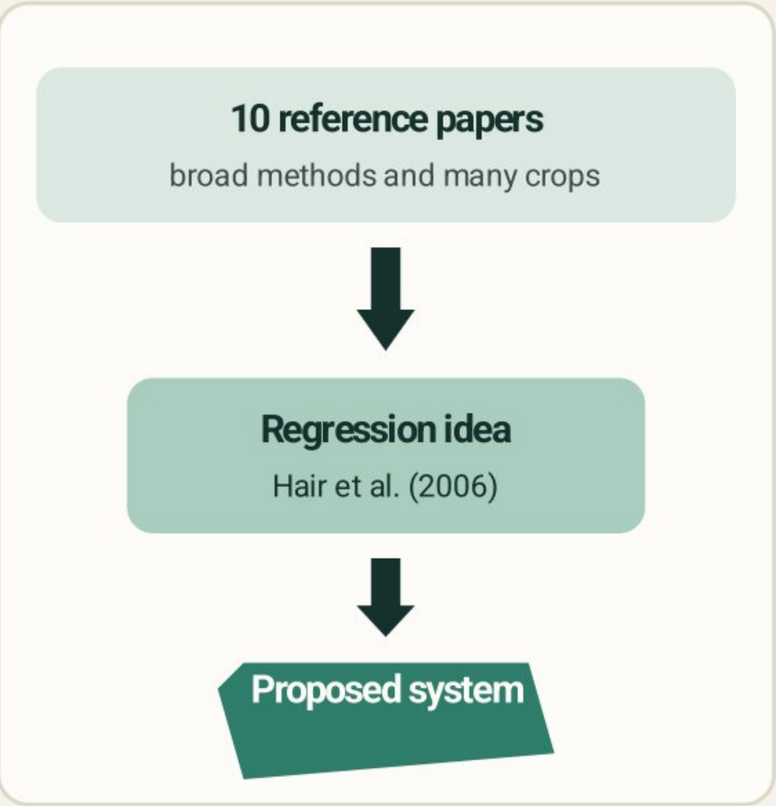
1. Mucherino et al. (2009): survey of data mining techniques in agriculture
2. Kokilavani & Geethalakshmi (2013): efficient rice, maize and groundnut cropping zones in Tamil Nadu
3. Chinchuluun & Xanthopoulos (2010): data mining in agricultural and environmental sciences
4. Suraparaju et al. (2014): machine learning to forecast crop yield from climatic parameters
5. Patel & Patel (2014): brief survey of data mining on agricultural data
6. Kumar & Kannathasan (2011): survey of data mining for soil data
7. Kalpana, Shanthi & Arumugam (2014): survey of data mining in agriculture
8. Mankar & Burange (2014): data mining, an evolutionary view of agriculture
9. Ramesh & Vishnu Vardhan (2013): data mining on agricultural yield data
10. Hair et al. (2006): Multivariate Data Analysis, 6th ed., the source for linear regression



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Reference Papers vs Proposed System

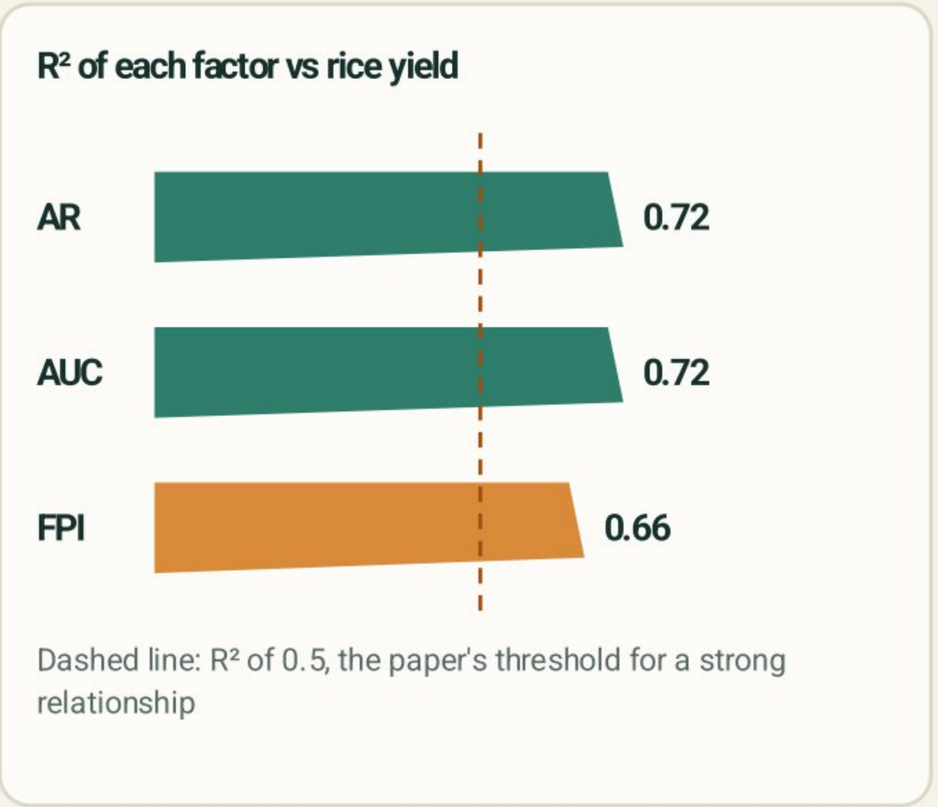
Aspect	Reference papers	Proposed system
Focus	Data mining in agriculture, cropping zones, climate-based forecasting	Rice yield against rainfall, area and food price
Methods	Classification, clustering, machine learning (k-means, SVM)	Linear regression, one factor at a time
Output	Surveys, zone indices, forecasts	R^2 for each factor
Ease of use	Often heavier models	Simple and easy to explain



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Existing vs Proposed System

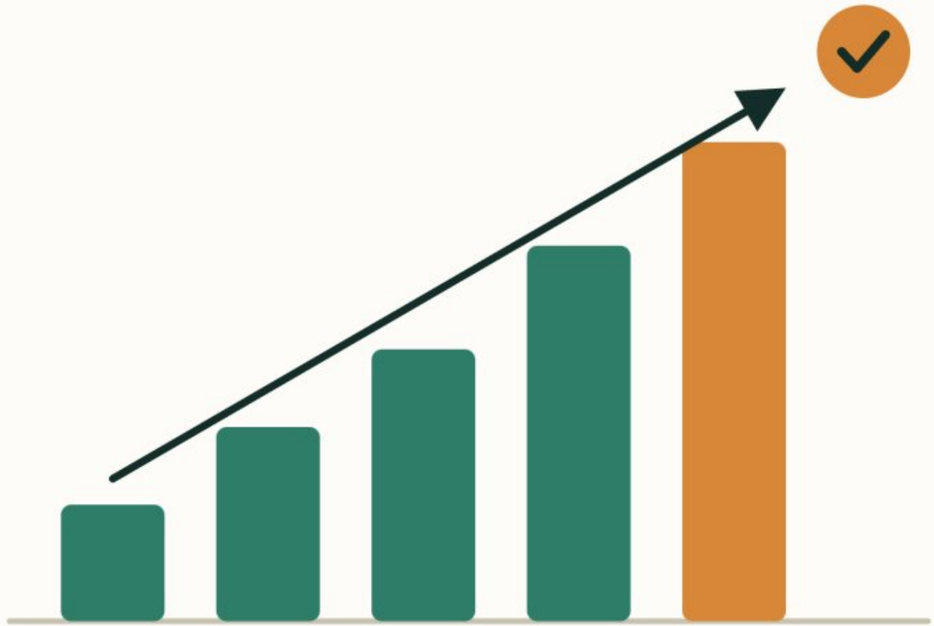
Aspect	Existing system	Proposed system
Basis	Experience, classification, zoning	Regression on AR, AUC, FPI
Factors	Examined in isolation	Rainfall, area and price together
Output	Descriptive, zone-level	Quantified with R^2 per factor
Insight	Harder to interpret	Easy to read and explain



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Advantages

- Simple linear model that is easy to explain.
- Measures each factor's strength with R^2: AR 0.72, AUC 0.72, FPI 0.66.
- Needs only public data from the Dept. of Statistics and Agriculture, NIC.
- Quick to run in MATLAB and easy to extend with more factors.



Each factor shows a clear link to yield

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Disadvantages

- Only three factors; soil, weather and support price are left out.
- Only rice is studied, so results may not carry over to other crops.
- A straight-line model can miss non-linear effects.
- R^2 shows association, not cause.

Included in this study

AR	AUC	FPI
Rainfall	Area	Food price

Not yet included

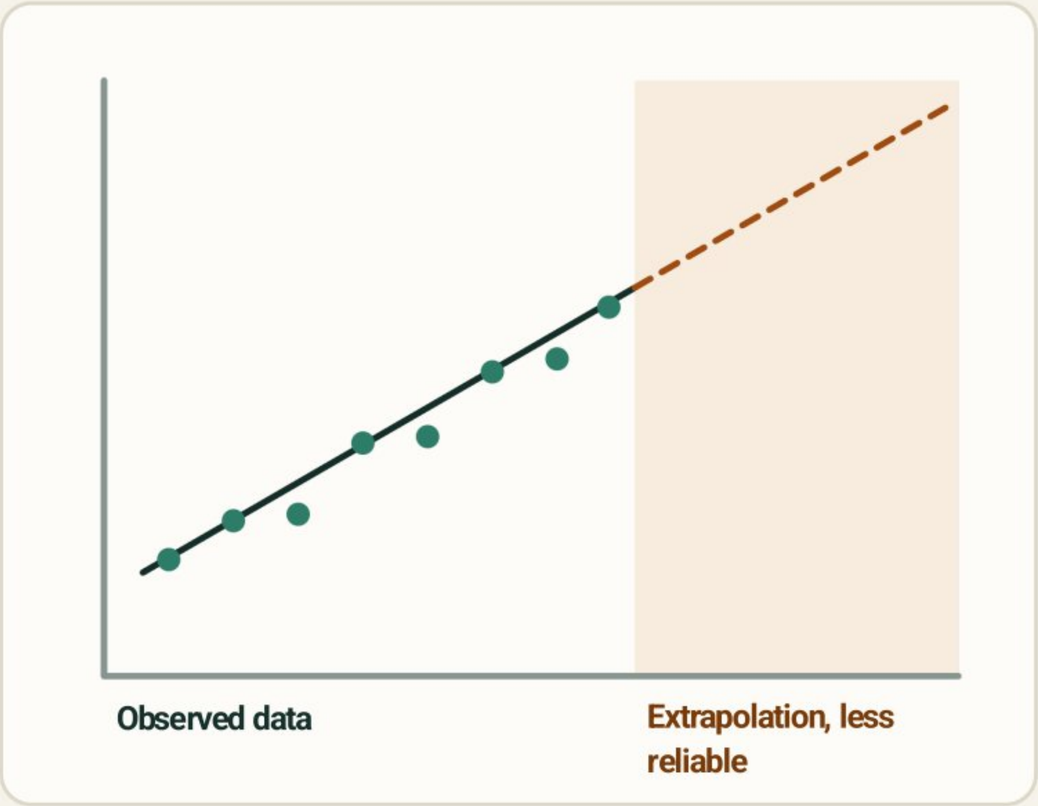
Soil	Weather	MSP
Parameters	Conditions	Support price

Future work named in the paper: MSP, weather conditions, soil parameters

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Limitations

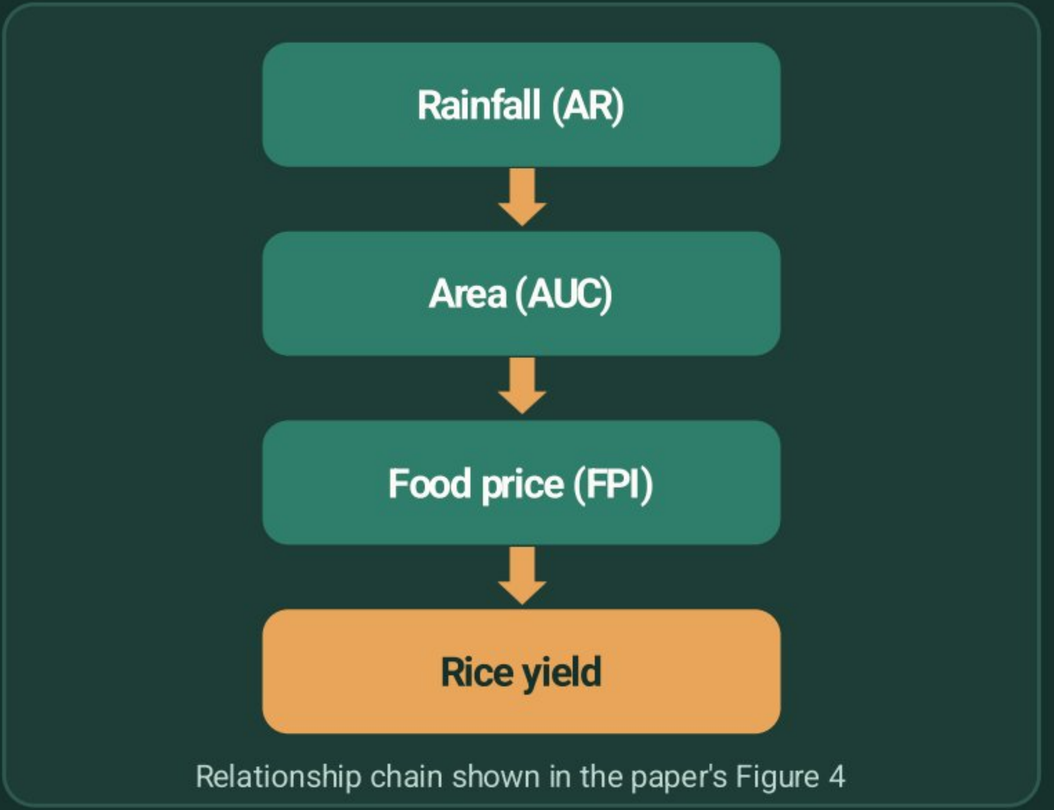
- Small sample: about ten years of rice data (Table 2).
- Irrigation figures are missing (NA) for 2008– 09 to 2010–11.
- R^2 is the only accuracy measure reported; no separate test data.
- Predicting outside the data range is extrapolation and less reliable.



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Conclusion

- Rainfall, area and food price index each show a strong link to rice yield (R^2 0.66 to 0.72).
- Rainfall stands out as the main driver, with area and price also influencing yield.
- Regression gives a simple, interpretable base for yield prediction.
- Next: add MSP, CPI, WPI, weather and soil factors.



Relationship chain shown in the paper's Figure 4

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